



856 Ship Notice/Manifest

X12: 856 Ship Notice/Manifest

Version: 2.018

Author:

SPS Commerce

Notes:

SPS Commerce can accommodate various versions. This specification is based on V5010. If you would like to use another version, please notify your SPS Commerce Solution Strategist.

Table of Contents

856 Ship Notice/Manifest	1
ISA Interchange Control Header	4
GS Functional Group Header	7
ST Transaction Set Header	9
BSN Beginning Segment for Ship Notice	10
DTM Date/Time Reference	12
HL Hierarchical Level	13
TD1 Carrier Details (Quantity and Weight)	14
TD5 Carrier Details (Routing Sequence/Transit Time)	16
TD3 Carrier Details (Equipment)	18
REF Reference Information	19
PER Administrative Communications Contact	20
DTM Date/Time Reference	22
FOB F.O.B. Related Instructions	23
N1 Party Identification	25
N2 Additional Name Information	27
N3 Party Location	28
N4 Geographic Location	29
HL Hierarchical Level	30
PRF Purchase Order Reference	31
TD1 Carrier Details (Quantity and Weight)	32
TD5 Carrier Details (Routing Sequence/Transit Time)	34
TD3 Carrier Details (Equipment)	36
REF Reference Information	37
DTM Date/Time Reference	38
N1 Party Identification	39
N2 Additional Name Information	41
N3 Party Location	42
N4 Geographic Location	43
HL Hierarchical Level	44
TD5 Carrier Details (Routing Sequence/Transit Time)	45
TD3 Carrier Details (Equipment)	47
REF Reference Information	48
PER Administrative Communications Contact	49
MAN Marks and Numbers Information	51
DTM Date/Time Reference	53
FOB F.O.B. Related Instructions	54
N1 Party Identification	55
N2 Additional Name Information	57

N3	Party Location	58
N4	Geographic Location	59
HL	Hierarchical Level	60
LIN	Item Identification	61
SN1	Item Detail (Shipment)	66
PO4	Item Physical Details	68
PID	Product/Item Description	71
REF	Reference Information	73
DTM	Date/Time Reference	74
CTT	Transaction Totals	75
SE	Transaction Set Trailer	76
GE	Functional Group Trailer	77
IEA	Interchange Control Trailer	78

856

Ship Notice/Manifest

Functional Group=SH

Purpose: This X12 Transaction Set contains the format and establishes the data contents of the Ship Notice/Manifest Transaction Set (856) for use within the context of an Electronic Data Interchange (EDI) environment. The transaction set can be used to list the contents of a shipment of goods as well as additional information relating to the shipment, such as order information, product description, physical characteristics, type of packaging, marking, carrier information, and configuration of goods within the transportation equipment. The transaction set enables the sender to describe the contents and configuration of a shipment in various levels of detail and provides an ordered flexibility to convey information. The sender of this transaction is the organization responsible for detailing and communicating the contents of a shipment, or shipments, to one or more receivers of the transaction set. The receiver of this transaction set can be any organization having an interest in the contents of a shipment or information about the contents of a shipment.

Not Defined:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
0100	ISA	Interchange Control Header	M	1			Must use
0200	GS	Functional Group Header	M	1			Must use

Heading:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
0100	ST	Transaction Set Header	M	1			Must use
0200	BSN	Beginning Segment for Ship Notice	M	1			Must use
0300	DTM	Date/Time Reference	O	10			Recommended

Detail:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
<u>LOOP ID - HL</u>					<u>200000</u>	<u>CN2/0100</u>	
						<u>L</u>	
0200	HL	Hierarchical Level	M	1		C2/0200	Must use
1200	TD1	Carrier Details (Quantity and Weight)	O	20			Recommended
1300	TD5	Carrier Details (Routing Sequence/Transit Time)	O	12			Recommended
<u>LOOP ID - TD3</u>					<u>12</u>		
1500	TD3	Carrier Details (Equipment)	O	1			Used
1900	REF	Reference Information	O	>1			Recommended
2000	PER	Administrative Communications Contact	O	3			Used
3700	DTM	Date/Time Reference	O	10			Used
3800	FOB	F.O.B. Related Instructions	O	1			Recommended
<u>LOOP ID - N1</u>					<u>200</u>		
4100	N1	Party Identification	O	1			Recommended
4200	N2	Additional Name Information	O	2			Used
4300	N3	Party Location	O	2			Recommended
4400	N4	Geographic Location	O	1			Recommended
<u>LOOP ID - HL</u>					<u>200000</u>	<u>CN2/0100</u>	
						<u>L</u>	

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
0200	HL	Hierarchical Level	M	1		C2/0200	Must use
0600	PRF	Purchase Order Reference	M	1			Must use
1200	TD1	Carrier Details (Quantity and Weight)	O	20			Recommended
1300	TD5	Carrier Details (Routing Sequence/Transit Time)	O	12			Recommended
<u>LOOP ID - TD3</u>				<u>12</u>			
1500	TD3	Carrier Details (Equipment)	O	1			Used
1900	REF	Reference Information	O	>1			Recommended
3700	DTM	Date/Time Reference	O	10			Used
<u>LOOP ID - N1</u>				<u>200</u>			
4100	N1	Party Identification	O	1			Recommended
4200	N2	Additional Name Information	O	2			Used
4300	N3	Party Location	O	2			Recommended
4400	N4	Geographic Location	O	1			Recommended
<u>LOOP ID - HL</u>				<u>200000</u>		<u>CN2/0100</u>	
						<u>L</u>	
0200	HL	Hierarchical Level	M	1		C2/0200	Must use
1300	TD5	Carrier Details (Routing Sequence/Transit Time)	O	12			Recommended
<u>LOOP ID - TD3</u>				<u>12</u>			
1500	TD3	Carrier Details (Equipment)	O	1			Used
1900	REF	Reference Information	O	>1			Recommended
2000	PER	Administrative Communications Contact	O	3			Used
3600	MAN	Marks and Numbers Information	O	>1			Used
3700	DTM	Date/Time Reference	O	10			Used
3800	FOB	F.O.B. Related Instructions	O	1			Recommended
<u>LOOP ID - N1</u>				<u>200</u>			
4100	N1	Party Identification	O	1			Recommended
4200	N2	Additional Name Information	O	2			Used
4300	N3	Party Location	O	2			Recommended
4400	N4	Geographic Location	O	1			Recommended
<u>LOOP ID - HL</u>				<u>200000</u>		<u>CN2/0100</u>	
						<u>L</u>	
0200	HL	Hierarchical Level	M	1		C2/0200	Must use
0300	LIN	Item Identification	M	1		N2/0300	Must use
0400	SN1	Item Detail (Shipment)	M	1			Must use
0700	PO4	Item Physical Details	O	1			Recommended
0800	PID	Product/Item Description	O	200			Recommended
1900	REF	Reference Information	O	>1			Recommended
3700	DTM	Date/Time Reference	O	10			Used

Summary:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
0100	CTT	Transaction Totals	O	1		N3/0100	Used
0200	SE	Transaction Set Trailer	M	1			Must use

Not Defined:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
0100	GE	Functional Group Trailer	M	1			Must use
0200	IEA	Interchange Control Trailer	M	1			Must use

Notes:

- 2/0100L S- Shipment
 2/0100L O- Order
 2/0100L T/P- Tare or Pack
 2/0100L I- Item
 2/0300 If an item qualifier (e.g. LIN02) is present, then the item ID (e.g. LIN03) is required, and vice versa.
 3/0100 Number of line items (CTT01) is the accumulation of the number of HL segments. If used, hash total (CTT02) is the sum of the value of units shipped (SN102) for each SN1 segment.

Comments:

- 2/0100L The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.
 2/0200 The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.
 2/0100L The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.
 2/0200 The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.
 2/0100L The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.
 2/0200 The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.
 2/0100L The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.
 2/0200 The HL segment is the only mandatory segment within the HL loop, and by itself, the HL segment has no meaning.

SPS Commerce Transaction Notes:

Element Data Types Key:

N: Used for numeric values. If decimals are implied, the number of digits after the implied decimal will show after N. (e.g. N2 means the value of 1234 would be interpreted as 12.34. N0 is the same as N)

R: Used for decimal values.

AN: Used for alphanumeric values

DT: Used for date values (format indicated in specification; CCYYMMDD)

TM: Used for time values (format indicated in specification; HHMMSS)

ID: Used for qualifiers, or code sets

ISA Interchange Control Header

Pos: 0100	Max: 1
Not Defined - Mandatory	
Loop: N/A	Elements: 16

User Option (Usage): Must use

Purpose: To start and identify an interchange of zero or more functional groups and interchange-related control segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
ISA01	I01	Authorization Information Qualifier Description: Code identifying the type of information in the Authorization Information CodeList Summary (Total Codes: 7, Included: 1) Code Name 00 No Authorization Information Present (No Meaningful Information in I02)	M	ID	2/2	Must use
ISA02	I02	Authorization Information Description: Information used for additional identification or authorization of the interchange sender or the data in the interchange; the type of information is set by the Authorization Information Qualifier (I01)	M	AN	10/10	Must use
ISA03	I03	Security Information Qualifier Description: Code identifying the type of information in the Security Information CodeList Summary (Total Codes: 2, Included: 1) Code Name 00 No Security Information Present (No Meaningful Information in I04)	M	ID	2/2	Must use
ISA04	I04	Security Information Description: This is used for identifying the security information about the interchange sender or the data in the interchange; the type of information is set by the Security Information Qualifier (I03)	M	AN	10/10	Must use
ISA05	I05	Interchange ID Qualifier Description: Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified All valid standard codes are used. (Total Codes: 41)	M	ID	2/2	Must use
ISA06	I06	Interchange Sender ID Description: Identification code published by the sender for other parties	M	AN	15/15	Must use

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		to use as the receiver ID to route data to them; the sender always codes this value in the sender ID element SPS Commerce Element Notes: <i>Let your SPS Commerce know how you will identify your trading partner(s). Please see the "Start Here Guide" for further information regarding Envelope IDs.</i>				
ISA07	I05	Interchange ID Qualifier Description: Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified All valid standard codes are used. (Total Codes: 41)	M	ID	2/2	Must use
ISA08	I07	Interchange Receiver ID Description: Identification code published by the receiver of the data; When sending, it is used by the sender as their sending ID, thus other parties sending to them will use this as a receiving ID to route data to them SPS Commerce Element Notes: <i>Let your SPS Commerce know how you will identify your trading partner(s). Please see the "Start Here Guide" for further information regarding Envelope IDs.</i>	M	AN	15/15	Must use
ISA09	I08	Interchange Date Description: Date of the interchange	M	DT	6/6	Must use
ISA10	I09	Interchange Time Description: Time of the interchange	M	TM	4/4	Must use
ISA11	I65	Repetition Separator Description: Type is not applicable; the repetition separator is a delimiter and not a data element; this field provides the delimiter used to separate repeated occurrences of a simple data element or a composite data structure; this value must be different than the data element separator, component element separator, and the segment terminator	M		1/1	Must use
ISA12	I11	Interchange Control Version Number Description: Code specifying the version number of the interchange control segments CodeList Summary (Total Codes: 20, Included: 3) <u>Code</u> <u>Name</u>	M	ID	5/5	Must use

		<u>Code</u>	<u>Name</u>				
		00401	Standards Approved for Publication by ASC X12 Procedures Review Board through October 1997				
		00403	Standards Approved for Publication by ASC X12 Procedures Review Board through October 1999				
		00501	Standards Approved for Publication by ASC X12 Procedures Review Board through October 2003				
ISA13	I12	Interchange Control Number		M	N0	9/9	Must use
		Description: A control number assigned by the interchange sender					
ISA14	I13	Acknowledgment Requested		M	ID	1/1	Must use
		Description: Code indicating sender's request for an interchange acknowledgment					
		All valid standard codes are used. (Total Codes: 2)					
ISA15	I14	Interchange Usage Indicator		M	ID	1/1	Must use
		Description: Code indicating whether data enclosed by this interchange envelope is test, production or information					
		CodeList Summary (Total Codes: 3, Included: 2)					
		<u>Code</u>	<u>Name</u>				
		P	Production Data				
		T	Test Data				
ISA16	I15	Component Element Separator		M		1/1	Must use
		Description: Type is not applicable; the component element separator is a delimiter and not a data element; this field provides the delimiter used to separate component data elements within a composite data structure; this value must be different than the data element separator and the segment terminator					

GS Functional Group Header

Pos: 0200	Max: 1
Not Defined - Mandatory	
Loop: N/A	Elements: 8

User Option (Usage): Must use

Purpose: To indicate the beginning of a functional group and to provide control information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
GS01	479	Functional Identifier Code	M	ID	2/2	Must use
		Description: Code identifying a group of application related transaction sets CodeList Summary (Total Codes: 260, Included: 1) <u>Code</u> <u>Name</u> SH Ship Notice/Manifest (856)				
GS02	142	Application Sender's Code	M	AN	2/15	Must use
		Description: Code identifying party sending transmission; codes agreed to by trading partners				
GS03	124	Application Receiver's Code	M	AN	2/15	Must use
		Description: Code identifying party receiving transmission; codes agreed to by trading partners				
GS04	373	Date	M	DT	8/8	Must use
		Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year				
GS05	337	Time	M	TM	4/8	Must use
		Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)				
GS06	28	Group Control Number	M	N0	1/9	Must use
		Description: Assigned number originated and maintained by the sender				
GS07	455	Responsible Agency Code	M	ID	1/2	Must use
		Description: Code identifying the issuer of the standard; this code is used in conjunction with Data Element 480 All valid standard codes are used. (Total Codes: 2)				

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
GS08	480	Version / Release / Industry Identifier Code	M	AN	1/12	Must use

Description: Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and subrelease, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed

CodeList Summary (Total Codes: 54, Included: 3)

<u>Code</u>	<u>Name</u>
004010	Standards Approved for Publication by ASC X12 Procedures Review Board through October 1997
004030	Standards Approved for Publication by ASC X12 Procedures Review Board through October 1999
005010	Standards Approved for Publication by ASC X12 Procedures Review Board through October 2003

Semantics:

1. GS04 is the group date.
2. GS05 is the group time.
3. The data interchange control number GS06 in this header must be identical to the same data element in the associated functional group trailer, GE02.

Comments:

1. A functional group of related transaction sets, within the scope of X12 standards, consists of a collection of similar transaction sets enclosed by a functional group header and a functional group trailer.

ST Transaction Set Header

Pos: 0100	Max: 1
Heading - Mandatory	
Loop: N/A	Elements: 3

User Option (Usage): Must use

Purpose: To indicate the start of a transaction set and to assign a control number

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
ST01	143	Transaction Set Identifier Code	M	ID	3/3	Must use
Description: Code uniquely identifying a Transaction Set CodeList Summary (Total Codes: 318, Included: 1) <u>Code</u> <u>Name</u> 856 Ship Notice/Manifest						
ST02	329	Transaction Set Control Number	M	AN	4/9	Must use
Description: Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set						
ST03	1705	Implementation Convention Reference	O	AN	1/35	Used
Description: Reference assigned to identify Implementation Convention						

Semantics:

1. The transaction set identifier (ST01) is used by the translation routines of the interchange partners to select the appropriate transaction set definition (e.g., 810 selects the Invoice Transaction Set).
2. The implementation convention reference (ST03) is used by the translation routines of the interchange partners to select the appropriate implementation convention to match the transaction set definition. When used, this implementation convention reference takes precedence over the implementation reference specified in the GS08.

BSN Beginning Segment for Ship Notice

Pos: 0200	Max: 1
Heading - Mandatory	
Loop: N/A	Elements: 5

User Option (Usage): Must use

Purpose: To transmit identifying numbers, dates, and other basic data relating to the transaction set

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
BSN01	353	Transaction Set Purpose Code	M	ID	2/2	Must use
Description: Code identifying purpose of transaction set SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> 00: Original 01: Cancellation 04: Change 05: Replace 06: Confirmation 07: Duplicate All valid standard codes are used. (Total Codes: 66)						
BSN02	396	Shipment Identification	M	AN	2/30	Must use
Description: A unique control number assigned by the original shipper to identify a specific shipment						
BSN03	373	Date	M	DT	8/8	Must use
Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year						
BSN04	337	Time	M	TM	4/8	Must use
Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)						
BSN05	1005	Hierarchical Structure Code	O	ID	4/4	Used
Description: Code indicating the hierarchical application structure of a transaction set that utilizes the HL segment to define the structure of the transaction set						

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For questions regarding shipment structure, please reach out to your Business Analyst.</i> <i>0001: Shipment, Order, Pack Item</i> All valid standard codes are used. (Total Codes: 81)				

Semantics:

1. BSN03 is the date the shipment transaction set is created.
2. BSN04 is the time the shipment transaction set is created.
3. BSN06 is limited to shipment related codes.

Comments:

1. BSN06 and BSN07 differentiate the functionality of use for the transaction set.

DTM Date/Time Reference

Pos: 0300	Max: 10
Heading - Optional	
Loop: N/A	Elements: 3

User Option (Usage): Recommended

Purpose: To specify pertinent dates and times

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
DTM01	374	Date/Time Qualifier	M	ID	3/3	Must use
		Description: Code specifying type of date or time, or both date and time SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> 001: Cancel Date 002: Requested Delivery Date 010: Requested Ship Date 011: Actual Ship Date 067: Current Schedule Delivery Date All valid standard codes are used. (Total Codes: 1280)				
DTM02	373	Date	X	DT	8/8	Used
		Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year				
DTM03	337	Time	X	TM	4/8	Used
		Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)				

Syntax Rules:

1. R020305 - At least one of DTM02, DTM03 or DTM05 is required.

HL Hierarchical Level

Pos: 0200	Max: 1
Detail - Mandatory	
Loop: HL	Elements: 3

User Option (Usage): Must use

Purpose: To identify dependencies among and the content of hierarchically related groups of data segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
HL01	628	Hierarchical ID Number Description: A unique number assigned by the sender to identify a particular data segment in a hierarchical structure	M	AN	1/12	Must use
HL02	734	Hierarchical Parent ID Number Description: Identification number of the next higher hierarchical data segment that the data segment being described is subordinate to	O	AN	1/12	Used
HL03	735	Hierarchical Level Code Description: Code defining the characteristic of a level in a hierarchical structure CodeList Summary (Total Codes: 250, Included: 1) <u>Code</u> <u>Name</u> S Shipment	M	ID	1/2	Must use

Comments:

1. The HL segment is used to identify levels of detail information using a hierarchical structure, such as relating line-item data to shipment data, and packaging data to line-item data.
2. The HL segment defines a top-down/left-right ordered structure.
3. HL01 shall contain a unique alphanumeric number for each occurrence of the HL segment in the transaction set. For example, HL01 could be used to indicate the number of occurrences of the HL segment, in which case the value of HL01 would be "1" for the initial HL segment and would be incremented by one in each subsequent HL segment within the transaction.
4. HL02 identifies the hierarchical ID number of the HL segment to which the current HL segment is subordinate.
5. HL03 indicates the context of the series of segments following the current HL segment up to the next occurrence of an HL segment in the transaction. For example, HL03 is used to indicate that subsequent segments in the HL loop form a logical grouping of data referring to shipment, order, or item-level information.
6. HL04 indicates whether or not there are subordinate (or child) HL segments related to the current HL segment.

TD1 Carrier Details (Quantity and Weight)

Pos: 1200	Max: 20
Detail - Optional	
Loop: HL	Elements: 8

User Option (Usage): Recommended

Purpose: To specify the transportation details relative to commodity, weight, and quantity

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD101	103	Packaging Code Description: Code identifying the type of packaging; Part 1: Packaging Form, Part 2: Packaging Material; if the Data Element is used, then Part 1 is always required SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CTN: Carton PLT: Pallet 25: Corrugated 94: Wood All valid standard codes are used. (Total Codes: 212)	O	AN	3/5	Recommended
TD102	80	Lading Quantity Description: Number of units (pieces) of the lading commodity	X	N0	1/7	Recommended
TD105	79	Lading Description Description: Description of an item as required for rating and billing purposes	O	AN	1/50	Used
TD106	187	Weight Qualifier Description: Code defining the type of weight SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> G: Gross Weight N: Actual Net Weight All valid standard codes are used. (Total Codes: 52)	O	ID	1/2	Recommended
TD107	81	Weight Description: Numeric value of weight	X	R	1/10	Recommended
TD108	355	Unit or Basis for Measurement Code Description: Code specifying the units in which a value is being expressed, or	X	ID	2/2	Recommended

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>LB: Pound</i> <i>KG: Kilogram</i>				
		All valid standard codes are used. (Total Codes: 844)				
TD109	183	Volume	X	R	1/8	Used
		Description: Value of volumetric measure				
TD110	355	Unit or Basis for Measurement Code	X	ID	2/2	Used
		Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>CF: Cubic Feet</i>				
		All valid standard codes are used. (Total Codes: 844)				

Syntax Rules:

1. C0102 - If TD101 is present, then TD102 is required.
2. C0607 - If TD106 is present, then TD107 is required.
3. P0708 - If either TD107 or TD108 is present, then the other is required.
4. P0910 - If either TD109 or TD110 is present, then the other is required.

TD5 Carrier Details (Routing Sequence/Transit Time)

Pos: 1300	Max: 12
Detail - Optional	
Loop: HL	Elements: 7

User Option (Usage): Recommended

Purpose: To specify the carrier and sequence of routing and provide transit time information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD501	133	Routing Sequence Code Description: Code describing the relationship of a carrier to a specific shipment movement SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> All valid standard codes are used. (Total Codes: 23)	O	ID	1/2	Used
TD502	66	Identification Code Qualifier Description: Code designating the system/method of code structure used for Identification Code (67) SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CodeList Summary (Total Codes: 241, Included: 1) Code Name 2 Standard Carrier Alpha Code (SCAC)	X	ID	1/2	Used
TD503	67	Identification Code Description: Code identifying a party or other code	X	AN	2/80	Used
TD504	91	Transportation Method/Type Code Description: Code specifying the method or type of transportation for the shipment SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>H: Customer Pickup</i> <i>M: Motor (Common Carrier)</i> <i>P: Private Parcel</i> <i>T: Best Way (Shipper's Option)</i> All valid standard codes are used. (Total Codes: 80)	X	ID	1/2	Recommended
TD505	387	Routing	X	AN	1/35	Recommended

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		Description: Free-form description of the routing or requested routing for shipment, or the originating carrier's identity				
TD506	368	Shipment/Order Status Code	X	ID	2/2	Used
		Description: Code indicating the status of an order or shipment or the disposition of any difference between the quantity ordered and the quantity shipped for a line item or transaction				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>CL: Complete Shipment</i> <i>PR: Partial Shipment</i>				
		All valid standard codes are used. (Total Codes: 102)				
TD512	284	Service Level Code	X	ID	2/2	Used
		Description: Code indicating the level of transportation service or the billing service offered by the transportation carrier				
		SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i>				
		All valid standard codes are used. (Total Codes: 71)				

Syntax Rules:

1. R0204050612 - At least one of TD502, TD504, TD505, TD506 or TD512 is required.
2. C0203 - If TD502 is present, then TD503 is required.
3. C0708 - If TD507 is present, then TD508 is required.
4. C1011 - If TD510 is present, then TD511 is required.

Semantics:

1. TD515 is the country where the service is to be performed.

Comments:

1. When specifying a routing sequence to be used for the shipment movement in lieu of specifying each carrier within the movement, use TD502 to identify the party responsible for defining the routing sequence, and use TD503 to identify the actual routing sequence, specified by the party identified in TD502.

TD3 Carrier Details (Equipment)

Pos: 1500	Max: 1
Detail - Optional	
Loop: TD3	Elements: 4

User Option (Usage): Used

Purpose: To specify transportation details relating to the equipment used by the carrier

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD301	40	Equipment Description Code	X	ID	2/2	Used
		Description: Code identifying type of equipment used for shipment SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. TL: Trailer (not otherwise specified)</i> All valid standard codes are used. (Total Codes: 139)				
TD302	206	Equipment Initial	O	AN	1/4	Used
		Description: Prefix or alphabetic part of an equipment unit's identifying number				
TD303	207	Equipment Number	X	AN	1/15	Used
		Description: Sequencing or serial part of an equipment unit's identifying number (pure numeric form for equipment number is preferred)				
TD309	225	Seal Number	O	AN	2/15	Used
		Description: Unique number on seal used to close a shipment				

Syntax Rules:

1. C0203 - If TD302 is present, then TD303 is required.

REF Reference Information

Pos: 1900	Max: >1
Detail - Optional	
Loop: HL	Elements: 2

User Option (Usage): Recommended

Purpose: To specify identifying information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
REF01	128	Reference Identification Qualifier	M	ID	2/3	Must use

Description: Code qualifying the Reference Identification

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

2I: Tracking Number

19: Division Number

IA: Vendor

CO: Customer Order Number

BM: Bill of Lading Number

CN: Carrier Pro Number

23: Trading Partner Identifier

VN: Vendor Order Number (Depositor Number)

**All valid standard codes are used.
(Total Codes: 1731)**

REF02	127	Reference Identification	X	AN	1/50	Used
-------	-----	--------------------------	---	----	------	------

Description: Reference information as defined for a particular Transaction Set or as specified by the Reference Identification Qualifier

Syntax Rules:

1. R0203 - At least one of REF02 or REF03 is required.

Semantics:

1. REF04 contains data relating to the value cited in REF02.

PER Administrative Communications Contact

Pos: 2000	Max: 3
Detail - Optional	
Loop: HL	Elements: 8

User Option (Usage): Used

Purpose: To identify a person or office to whom administrative communications should be directed

Element Summary:

Ref	Id	Element Name	Req	Type	Min/Max	Usage
PER01	366	Contact Function Code	M	ID	2/2	Must use

Description: Code identifying the major duty or responsibility of the person or group named

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

IC: Information Contact

BD: Buyer Name or Buyer's Department

OC: Order Contact

**All valid standard codes are used.
(Total Codes: 238)**

PER02	93	Name	O	AN	1/60	Used
-------	----	------	---	----	------	------

Description: Free-form name

PER03	365	Communication Number Qualifier	X	ID	2/2	Used
-------	-----	--------------------------------	---	----	-----	------

Description: Code identifying the type of communication number

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

AP: Alternate Phone Number

EM: Electronic Mail

TE: Telephone Number

FX: Fax Number

**All valid standard codes are used.
(Total Codes: 42)**

PER04	364	Communication Number	X	AN	1/256	Used
-------	-----	----------------------	---	----	-------	------

Description: Complete communications number including country or area code when applicable

PER05	365	Communication Number Qualifier	X	ID	2/2	Used
-------	-----	--------------------------------	---	----	-----	------

Description: Code identifying the type of communication number

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		<i>specific to your trading partners, please reach out to your Business Analyst. AP: Alternate Phone Number EM: Electronic Mail TE: Telephone Number FX: Fax Number</i>				
		All valid standard codes are used. (Total Codes: 42)				
PER06	364	Communication Number	X	AN	1/256	Used
		Description: Complete communications number including country or area code when applicable				
PER07	365	Communication Number Qualifier	X	ID	2/2	Used
		Description: Code identifying the type of communication number				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. AP: Alternate Phone Number EM: Electronic Mail TE: Telephone Number FX: Fax Number</i>				
		All valid standard codes are used. (Total Codes: 42)				
PER08	364	Communication Number	X	AN	1/256	Used
		Description: Complete communications number including country or area code when applicable				

Syntax Rules:

1. P0304 - If either PER03 or PER04 is present, then the other is required.
2. P0506 - If either PER05 or PER06 is present, then the other is required.
3. P0708 - If either PER07 or PER08 is present, then the other is required.

DTM Date/Time Reference

Pos: 3700	Max: 10
Detail - Optional	
Loop: HL	Elements: 3

User Option (Usage): Used

Purpose: To specify pertinent dates and times

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
DTM01	374	Date/Time Qualifier	M	ID	3/3	Must use
Description: Code specifying type of date or time, or both date and time SPS Commerce Element Notes: For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. All valid standard codes are used. (Total Codes: 1280)						
DTM02	373	Date	X	DT	8/8	Used
Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year						
DTM03	337	Time	X	TM	4/8	Used
Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)						

Syntax Rules:

1. R020305 - At least one of DTM02, DTM03 or DTM05 is required.

FOB F.O.B. Related Instructions

Pos: 3800	Max: 1
Detail - Optional	
Loop: HL	Elements: 5

User Option (Usage): Recommended

Purpose: To specify transportation instructions relating to shipment

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
FOB01	146	Shipment Method of Payment	M	ID	2/2	Must use
		Description: Code identifying payment terms for transportation charges SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CC: Collect PP: Prepaid (by seller) DF: Defined by Buyer and Seller All valid standard codes are used. (Total Codes: 31)				
FOB02	309	Location Qualifier	X	ID	1/2	Recommended
		Description: Code identifying type of location SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> OR: Origin DE: Destination All valid standard codes are used. (Total Codes: 184)				
FOB03	352	Description	O	AN	1/80	Recommended
		Description: A free-form description to clarify the related data elements and their content				
FOB04	334	Transportation Terms Qualifier Code	O	ID	2/2	Used
		Description: Code identifying the source of the transportation terms SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> All valid standard codes are used. (Total Codes: 3)				
FOB05	335	Transportation Terms Code	X	ID	3/3	Used
		Description: Code identifying the trade				

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		terms which apply to the shipment transportation responsibility				
		SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i>				
		All valid standard codes are used. (Total Codes: 24)				

Syntax Rules:

1. C0302 - If FOB03 is present, then FOB02 is required.
2. C0405 - If FOB04 is present, then FOB05 is required.
3. C0706 - If FOB07 is present, then FOB06 is required.
4. C0809 - If FOB08 is present, then FOB09 is required.

Semantics:

1. FOB01 indicates which party will pay the carrier.
2. FOB02 is the code specifying transportation responsibility location.
3. FOB06 is the code specifying the title passage location.
4. FOB08 is the code specifying the point at which the risk of loss transfers. This may be different than the location specified in FOB02/FOB03 and FOB06/FOB07.

N1 Party Identification

Pos: 4100	Max: 1
Detail - Optional	
Loop: N1	Elements: 4

User Option (Usage): Recommended

Purpose: To identify a party by type of organization, name, and code

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N101	98	Entity Identifier Code	M	ID	2/3	Must use
		Description: Code identifying an organizational entity, a physical location, property or an individual SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> ST: Ship To SF: Ship From VN: Vendor BT: Bill To Z7: Mark For Party All valid standard codes are used. (Total Codes: 1500)				
N102	93	Name	X	AN	1/60	Used
		Description: Free-form name				
N103	66	Identification Code Qualifier	X	ID	1/2	Used
		Description: Code designating the system/method of code structure used for Identification Code (67) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> 1: DUNS 9: DUNS + 4 91: Assigned by Seller 92: Assigned by Buyer All valid standard codes are used. (Total Codes: 241)				
N104	67	Identification Code	X	AN	2/80	Used
		Description: Code identifying a party or other code				

Syntax Rules:

1. R0203 - At least one of N102 or N103 is required.
2. P0304 - If either N103 or N104 is present, then the other is required.

Comments:

1. This segment, used alone, provides the most efficient method of providing organizational identification. To obtain this efficiency the "ID Code" (N104) must provide a key to the table maintained by the transaction processing party.
2. N105 and N106 further define the type of entity in N101.

N2 Additional Name Information

Pos: 4200	Max: 2
Detail - Optional	
Loop: N1	Elements: 2

User Option (Usage): Used

Purpose: To specify additional names

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N201	93	Name	M	AN	1/60	Must use
		Description: Free-form name				
N202	93	Name	O	AN	1/60	Used
		Description: Free-form name				

N3 Party Location

Pos: 4300	Max: 2
Detail - Optional	
Loop: N1	Elements: 2

User Option (Usage): Recommended

Purpose: To specify the location of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N301	166	Address Information	M	AN	1/55	Must use
		Description: Address information				
N302	166	Address Information	O	AN	1/55	Used
		Description: Address information				

N4 Geographic Location

Pos: 4400	Max: 1
Detail - Optional	
Loop: N1	Elements: 7

User Option (Usage): Recommended

Purpose: To specify the geographic place of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N401	19	City Name	O	AN	2/30	Used
		Description: Free-form text for city name				
N402	156	State or Province Code	X	ID	2/2	Used
		Description: Code (Standard State/Province) as defined by appropriate government agency				
N403	116	Postal Code	O	ID	3/15	Used
		Description: Code defining international postal zone code excluding punctuation and blanks (zip code for United States)				
N404	26	Country Code	X	ID	2/3	Used
		Description: Code identifying the country				
N405	309	Location Qualifier	X	ID	1/2	Used
		Description: Code identifying type of location				
		All valid standard codes are used. (Total Codes: 184)				
N406	310	Location Identifier	O	AN	1/30	Used
		Description: Code which identifies a specific location				
N407	1715	Country Subdivision Code	X	ID	1/3	Used
		Description: Code identifying the country subdivision				

Syntax Rules:

1. E0207 - Only one of N402 or N407 may be present.
2. C0605 - If N406 is present, then N405 is required.
3. C0704 - If N407 is present, then N404 is required.

Comments:

1. A combination of either N401 through N404, or N405 and N406 may be adequate to specify a location.
2. N402 is required only if city name (N401) is in the U.S. or Canada.

HL Hierarchical Level

Pos: 0200	Max: 1
Detail - Mandatory	
Loop: HL	Elements: 3

User Option (Usage): Must use

Purpose: To identify dependencies among and the content of hierarchically related groups of data segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
HL01	628	Hierarchical ID Number	M	AN	1/12	Must use
		Description: A unique number assigned by the sender to identify a particular data segment in a hierarchical structure				
HL02	734	Hierarchical Parent ID Number	O	AN	1/12	Used
		Description: Identification number of the next higher hierarchical data segment that the data segment being described is subordinate to				
HL03	735	Hierarchical Level Code	M	ID	1/2	Must use
		Description: Code defining the characteristic of a level in a hierarchical structure				
		CodeList Summary (Total Codes: 250, Included: 1)				
		<u>Code</u>	<u>Name</u>			
		O	Order			

Comments:

1. The HL segment is used to identify levels of detail information using a hierarchical structure, such as relating line-item data to shipment data, and packaging data to line-item data.
2. The HL segment defines a top-down/left-right ordered structure.
3. HL01 shall contain a unique alphanumeric number for each occurrence of the HL segment in the transaction set. For example, HL01 could be used to indicate the number of occurrences of the HL segment, in which case the value of HL01 would be "1" for the initial HL segment and would be incremented by one in each subsequent HL segment within the transaction.
4. HL02 identifies the hierarchical ID number of the HL segment to which the current HL segment is subordinate.
5. HL03 indicates the context of the series of segments following the current HL segment up to the next occurrence of an HL segment in the transaction. For example, HL03 is used to indicate that subsequent segments in the HL loop form a logical grouping of data referring to shipment, order, or item-level information.
6. HL04 indicates whether or not there are subordinate (or child) HL segments related to the current HL segment.

PRF Purchase Order Reference

Pos: 0600	Max: 1
Detail - Mandatory	
Loop: HL	Elements: 2

User Option (Usage): Must use

Purpose: To provide reference to a specific purchase order

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
PRF01	324	Purchase Order Number Description: Identifying number for Purchase Order assigned by the orderer/purchaser	M	AN	1/22	Must use
PRF04	373	Date Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year	O	DT	8/8	Recommended

Semantics:

1. PRF04 is the date assigned by the purchaser to purchase order.

TD1 Carrier Details (Quantity and Weight)

Pos: 1200	Max: 20
Detail - Optional	
Loop: HL	Elements: 8

User Option (Usage): Recommended

Purpose: To specify the transportation details relative to commodity, weight, and quantity

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD101	103	Packaging Code Description: Code identifying the type of packaging; Part 1: Packaging Form, Part 2: Packaging Material; if the Data Element is used, then Part 1 is always required SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CTN: Carton PLT: Pallet 25: Corrugated 94: Wood All valid standard codes are used. (Total Codes: 212)	O	AN	3/5	Recommended
TD102	80	Lading Quantity Description: Number of units (pieces) of the lading commodity	X	N0	1/7	Recommended
TD105	79	Lading Description Description: Description of an item as required for rating and billing purposes	O	AN	1/50	Used
TD106	187	Weight Qualifier Description: Code defining the type of weight SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> G: Gross Weight N: Actual Net Weight All valid standard codes are used. (Total Codes: 52)	O	ID	1/2	Recommended
TD107	81	Weight Description: Numeric value of weight	X	R	1/10	Recommended
TD108	355	Unit or Basis for Measurement Code Description: Code specifying the units in which a value is being expressed, or	X	ID	2/2	Recommended

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>LB: Pound</i> <i>KG: Kilogram</i>				
		All valid standard codes are used. (Total Codes: 844)				
TD109	183	Volume	X	R	1/8	Used
		Description: Value of volumetric measure				
TD110	355	Unit or Basis for Measurement Code	X	ID	2/2	Used
		Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>CF: Cubic Feet</i>				
		All valid standard codes are used. (Total Codes: 844)				

Syntax Rules:

1. C0102 - If TD101 is present, then TD102 is required.
2. C0607 - If TD106 is present, then TD107 is required.
3. P0708 - If either TD107 or TD108 is present, then the other is required.
4. P0910 - If either TD109 or TD110 is present, then the other is required.

TD5 Carrier Details (Routing Sequence/Transit Time)

Pos: 1300	Max: 12
Detail - Optional	
Loop: HL	Elements: 7

User Option (Usage): Recommended

Purpose: To specify the carrier and sequence of routing and provide transit time information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD501	133	Routing Sequence Code	O	ID	1/2	Used
Description: Code describing the relationship of a carrier to a specific shipment movement SPS Commerce Element Notes: For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. All valid standard codes are used. (Total Codes: 23)						
TD502	66	Identification Code Qualifier	X	ID	1/2	Used
Description: Code designating the system/method of code structure used for Identification Code (67) SPS Commerce Element Notes: For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. CodeList Summary (Total Codes: 241, Included: 1) <u>Code</u> <u>Name</u> 2 Standard Carrier Alpha Code (SCAC)						
TD503	67	Identification Code	X	AN	2/80	Used
Description: Code identifying a party or other code						
TD504	91	Transportation Method/Type Code	X	ID	1/2	Recommended
Description: Code specifying the method or type of transportation for the shipment SPS Commerce Element Notes: Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. H: Customer Pickup M: Motor (Common Carrier) P: Private Parcel T: Best Way (Shipper's Option) All valid standard codes are used. (Total Codes: 80)						
TD505	387	Routing	X	AN	1/35	Recommended

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		Description: Free-form description of the routing or requested routing for shipment, or the originating carrier's identity				
TD506	368	Shipment/Order Status Code	X	ID	2/2	Used
		Description: Code indicating the status of an order or shipment or the disposition of any difference between the quantity ordered and the quantity shipped for a line item or transaction				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>CL: Complete Shipment</i> <i>PR: Partial Shipment</i>				
		All valid standard codes are used. (Total Codes: 102)				
TD512	284	Service Level Code	X	ID	2/2	Used
		Description: Code indicating the level of transportation service or the billing service offered by the transportation carrier				
		SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i>				
		All valid standard codes are used. (Total Codes: 71)				

Syntax Rules:

1. R0204050612 - At least one of TD502, TD504, TD505, TD506 or TD512 is required.
2. C0203 - If TD502 is present, then TD503 is required.
3. C0708 - If TD507 is present, then TD508 is required.
4. C1011 - If TD510 is present, then TD511 is required.

Semantics:

1. TD515 is the country where the service is to be performed.

Comments:

1. When specifying a routing sequence to be used for the shipment movement in lieu of specifying each carrier within the movement, use TD502 to identify the party responsible for defining the routing sequence, and use TD503 to identify the actual routing sequence, specified by the party identified in TD502.

TD3 Carrier Details (Equipment)

Pos: 1500	Max: 1
Detail - Optional	
Loop: TD3	Elements: 4

User Option (Usage): Used

Purpose: To specify transportation details relating to the equipment used by the carrier

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD301	40	Equipment Description Code	X	ID	2/2	Used
		Description: Code identifying type of equipment used for shipment SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. TL: Trailer (not otherwise specified)</i> All valid standard codes are used. (Total Codes: 139)				
TD302	206	Equipment Initial	O	AN	1/4	Used
		Description: Prefix or alphabetic part of an equipment unit's identifying number				
TD303	207	Equipment Number	X	AN	1/15	Used
		Description: Sequencing or serial part of an equipment unit's identifying number (pure numeric form for equipment number is preferred)				
TD309	225	Seal Number	O	AN	2/15	Used
		Description: Unique number on seal used to close a shipment				

Syntax Rules:

1. E0110 - Only one of TD301 or TD310 may be present.
2. C0203 - If TD302 is present, then TD303 is required.
3. C0405 - If TD304 is present, then TD305 is required.
4. P0506 - If either TD305 or TD306 is present, then the other is required.

REF Reference Information

Pos: 1900	Max: >1
Detail - Optional	
Loop: HL	Elements: 2

User Option (Usage): Recommended

Purpose: To specify identifying information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
REF01	128	Reference Identification Qualifier	M	ID	2/3	Must use

Description: Code qualifying the Reference Identification

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

2I: Tracking Number

19: Division Number

IA: Vendor

CO: Customer Order Number

BM: Bill of Lading Number

CN: Carrier Pro Number

VN: Vendor Order Number (Depositor Number)

23: Trading Partner Identifier

**All valid standard codes are used.
(Total Codes: 1731)**

REF02	127	Reference Identification	X	AN	1/50	Used
-------	-----	--------------------------	---	----	------	------

Description: Reference information as defined for a particular Transaction Set or as specified by the Reference Identification Qualifier

Syntax Rules:

1. R0203 - At least one of REF02 or REF03 is required.

Semantics:

1. REF04 contains data relating to the value cited in REF02.

DTM Date/Time Reference

Pos: 3700	Max: 10
Detail - Optional	
Loop: HL	Elements: 3

User Option (Usage): Used

Purpose: To specify pertinent dates and times

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
DTM01	374	Date/Time Qualifier	M	ID	3/3	Must use
		Description: Code specifying type of date or time, or both date and time SPS Commerce Element Notes: For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. All valid standard codes are used. (Total Codes: 1280)				
DTM02	373	Date	X	DT	8/8	Used
		Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year				
DTM03	337	Time	X	TM	4/8	Used
		Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)				

Syntax Rules:

1. R020305 - At least one of DTM02, DTM03 or DTM05 is required.

N1 Party Identification

Pos: 4100	Max: 1
Detail - Optional	
Loop: N1	Elements: 4

User Option (Usage): Recommended

Purpose: To identify a party by type of organization, name, and code

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N101	98	Entity Identifier Code	M	ID	2/3	Must use
		Description: Code identifying an organizational entity, a physical location, property or an individual SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below, but a list of qualifiers specific to your trading partners will be provided by your Solution Strategist during the design phase.</i> BY: Buying Party Z7: Mark For Party All valid standard codes are used. (Total Codes: 1500)				
N102	93	Name	X	AN	1/60	Used
		Description: Free-form name				
N103	66	Identification Code Qualifier	X	ID	1/2	Used
		Description: Code designating the system/method of code structure used for Identification Code (67) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below, but a list of qualifiers specific to your trading partners will be provided by your Solution Strategist during the design phase.</i> 1: DUNS 9: DUNS + 4 91: Assigned by Seller 92: Assigned by Buyer All valid standard codes are used. (Total Codes: 241)				
N104	67	Identification Code	X	AN	2/80	Used
		Description: Code identifying a party or other code				

Syntax Rules:

1. R0203 - At least one of N102 or N103 is required.
2. P0304 - If either N103 or N104 is present, then the other is required.

Comments:

1. This segment, used alone, provides the most efficient method of providing organizational identification. To obtain this efficiency the "ID Code" (N104) must provide a key to the table maintained by the transaction processing party.
2. N105 and N106 further define the type of entity in N101.

SPS Commerce Segment Notes:

For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

N2 Additional Name Information

Pos: 4200	Max: 2
Detail - Optional	
Loop: N1	Elements: 2

User Option (Usage): Used

Purpose: To specify additional names

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N201	93	Name	M	AN	1/60	Must use
		Description: Free-form name				
N202	93	Name	O	AN	1/60	Used
		Description: Free-form name				

N3 Party Location

Pos: 4300	Max: 2
Detail - Optional	
Loop: N1	Elements: 2

User Option (Usage): Recommended

Purpose: To specify the location of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N301	166	Address Information	M	AN	1/55	Must use
		Description: Address information				
N302	166	Address Information	O	AN	1/55	Used
		Description: Address information				

N4 Geographic Location

Pos: 4400	Max: 1
Detail - Optional	
Loop: N1	Elements: 7

User Option (Usage): Recommended

Purpose: To specify the geographic place of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N401	19	City Name	O	AN	2/30	Used
		Description: Free-form text for city name				
N402	156	State or Province Code	X	ID	2/2	Used
		Description: Code (Standard State/Province) as defined by appropriate government agency				
N403	116	Postal Code	O	ID	3/15	Used
		Description: Code defining international postal zone code excluding punctuation and blanks (zip code for United States)				
N404	26	Country Code	X	ID	2/3	Used
		Description: Code identifying the country				
N405	309	Location Qualifier	X	ID	1/2	Used
		Description: Code identifying type of location				
		All valid standard codes are used. (Total Codes: 184)				
N406	310	Location Identifier	O	AN	1/30	Used
		Description: Code which identifies a specific location				
N407	1715	Country Subdivision Code	X	ID	1/3	Used
		Description: Code identifying the country subdivision				

Syntax Rules:

1. E0207 - Only one of N402 or N407 may be present.
2. C0605 - If N406 is present, then N405 is required.
3. C0704 - If N407 is present, then N404 is required.

Comments:

1. A combination of either N401 through N404, or N405 and N406 may be adequate to specify a location.
2. N402 is required only if city name (N401) is in the U.S. or Canada.

HL Hierarchical Level

Pos: 0200	Max: 1
Detail - Mandatory	
Loop: HL	Elements: 3

User Option (Usage): Must use

Purpose: To identify dependencies among and the content of hierarchically related groups of data segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>						
HL01	628	Hierarchical ID Number Description: A unique number assigned by the sender to identify a particular data segment in a hierarchical structure	M	AN	1/12	Must use						
HL02	734	Hierarchical Parent ID Number Description: Identification number of the next higher hierarchical data segment that the data segment being described is subordinate to	O	AN	1/12	Used						
HL03	735	Hierarchical Level Code Description: Code defining the characteristic of a level in a hierarchical structure CodeList Summary (Total Codes: 250, Included: 2) <table><tr><th><u>Code</u></th><th><u>Name</u></th></tr><tr><td>P</td><td>Pack</td></tr><tr><td>T</td><td>Shipping Tare</td></tr></table>	<u>Code</u>	<u>Name</u>	P	Pack	T	Shipping Tare	M	ID	1/2	Must use
<u>Code</u>	<u>Name</u>											
P	Pack											
T	Shipping Tare											

Comments:

1. The HL segment is used to identify levels of detail information using a hierarchical structure, such as relating line-item data to shipment data, and packaging data to line-item data.
2. The HL segment defines a top-down/left-right ordered structure.
3. HL01 shall contain a unique alphanumeric number for each occurrence of the HL segment in the transaction set. For example, HL01 could be used to indicate the number of occurrences of the HL segment, in which case the value of HL01 would be "1" for the initial HL segment and would be incremented by one in each subsequent HL segment within the transaction.
4. HL02 identifies the hierarchical ID number of the HL segment to which the current HL segment is subordinate.
5. HL03 indicates the context of the series of segments following the current HL segment up to the next occurrence of an HL segment in the transaction. For example, HL03 is used to indicate that subsequent segments in the HL loop form a logical grouping of data referring to shipment, order, or item-level information.
6. HL04 indicates whether or not there are subordinate (or child) HL segments related to the current HL segment.

TD5 Carrier Details (Routing Sequence/Transit Time)

Pos: 1300	Max: 12
Detail - Optional	
Loop: HL	Elements: 7

User Option (Usage): Recommended

Purpose: To specify the carrier and sequence of routing and provide transit time information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD501	133	Routing Sequence Code Description: Code describing the relationship of a carrier to a specific shipment movement SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> All valid standard codes are used. (Total Codes: 23)	O	ID	1/2	Used
TD502	66	Identification Code Qualifier Description: Code designating the system/method of code structure used for Identification Code (67) SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CodeList Summary (Total Codes: 241, Included: 1) Code Name 2 Standard Carrier Alpha Code (SCAC)	X	ID	1/2	Used
TD503	67	Identification Code Description: Code identifying a party or other code	X	AN	2/80	Used
TD504	91	Transportation Method/Type Code Description: Code specifying the method or type of transportation for the shipment SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>H: Customer Pickup</i> <i>M: Motor (Common Carrier)</i> <i>P: Private Parcel</i> <i>T: Best Way (Shipper's Option)</i> All valid standard codes are used. (Total Codes: 80)	X	ID	1/2	Recommended
TD505	387	Routing	X	AN	1/35	Recommended

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		Description: Free-form description of the routing or requested routing for shipment, or the originating carrier's identity				
TD506	368	Shipment/Order Status Code	X	ID	2/2	Used
		Description: Code indicating the status of an order or shipment or the disposition of any difference between the quantity ordered and the quantity shipped for a line item or transaction				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>CL: Complete Shipment</i> <i>PR: Partial Shipment</i>				
		All valid standard codes are used. (Total Codes: 102)				
TD512	284	Service Level Code	X	ID	2/2	Used
		Description: Code indicating the level of transportation service or the billing service offered by the transportation carrier				
		SPS Commerce Element Notes: <i>For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i>				
		All valid standard codes are used. (Total Codes: 71)				

Syntax Rules:

1. R0204050612 - At least one of TD502, TD504, TD505, TD506 or TD512 is required.
2. C0203 - If TD502 is present, then TD503 is required.
3. C0708 - If TD507 is present, then TD508 is required.
4. C1011 - If TD510 is present, then TD511 is required.

Semantics:

1. TD515 is the country where the service is to be performed.

Comments:

1. When specifying a routing sequence to be used for the shipment movement in lieu of specifying each carrier within the movement, use TD502 to identify the party responsible for defining the routing sequence, and use TD503 to identify the actual routing sequence, specified by the party identified in TD502.

TD3 Carrier Details (Equipment)

Pos: 1500	Max: 1
Detail - Optional	
Loop: TD3	Elements: 4

User Option (Usage): Used

Purpose: To specify transportation details relating to the equipment used by the carrier

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TD301	40	Equipment Description Code	X	ID	2/2	Used
		Description: Code identifying type of equipment used for shipment SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. TL: Trailer (not otherwise specified)</i> All valid standard codes are used. (Total Codes: 139)				
TD302	206	Equipment Initial	O	AN	1/4	Used
		Description: Prefix or alphabetic part of an equipment unit's identifying number				
TD303	207	Equipment Number	X	AN	1/15	Used
		Description: Sequencing or serial part of an equipment unit's identifying number (pure numeric form for equipment number is preferred)				
TD309	225	Seal Number	O	AN	2/15	Used
		Description: Unique number on seal used to close a shipment				

Syntax Rules:

1. E0110 - Only one of TD301 or TD310 may be present.
2. C0203 - If TD302 is present, then TD303 is required.
3. C0405 - If TD304 is present, then TD305 is required.
4. P0506 - If either TD305 or TD306 is present, then the other is required.

REF Reference Information

Pos: 1900	Max: >1
Detail - Optional	
Loop: HL	Elements: 2

User Option (Usage): Recommended

Purpose: To specify identifying information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
REF01	128	Reference Identification Qualifier	M	ID	2/3	Must use
Description: Code qualifying the Reference Identification SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> 2I: Tracking Number 19: Division Number IA: Vendor CO: Customer Order Number BM: Bill of Lading Number CN: Carrier Pro Number All valid standard codes are used. (Total Codes: 1731)						
REF02	127	Reference Identification	X	AN	1/50	Used
Description: Reference information as defined for a particular Transaction Set or as specified by the Reference Identification Qualifier						

Syntax Rules:

1. R0203 - At least one of REF02 or REF03 is required.

Semantics:

1. REF04 contains data relating to the value cited in REF02.

PER Administrative Communications Contact

Pos: 2000	Max: 3
Detail - Optional	
Loop: HL	Elements: 8

User Option (Usage): Used

Purpose: To identify a person or office to whom administrative communications should be directed

Element Summary:

Ref	Id	Element Name	Req	Type	Min/Max	Usage
PER01	366	Contact Function Code Description: Code identifying the major duty or responsibility of the person or group named SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>IC: Information Contact</i> <i>BD: Buyer Name or Buyer's Department</i> <i>OC: Order Contact</i> All valid standard codes are used. (Total Codes: 238)	M	ID	2/2	Must use
PER02	93	Name Description: Free-form name	O	AN	1/60	Used
PER03	365	Communication Number Qualifier Description: Code identifying the type of communication number SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>AP: Alternate Phone Number</i> <i>EM: Electronic Mail</i> <i>TE: Telephone Number</i> <i>FX: Fax Number</i> All valid standard codes are used. (Total Codes: 42)	X	ID	2/2	Used
PER04	364	Communication Number Description: Complete communications number including country or area code when applicable	X	AN	1/256	Used
PER05	365	Communication Number Qualifier Description: Code identifying the type of communication number SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers</i>	X	ID	2/2	Used

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		<i>specific to your trading partners, please reach out to your Business Analyst. AP: Alternate Phone Number EM: Electronic Mail TE: Telephone Number FX: Fax Number</i>				
		All valid standard codes are used. (Total Codes: 42)				
PER06	364	Communication Number	X	AN	1/256	Used
		Description: Complete communications number including country or area code when applicable				
PER07	365	Communication Number Qualifier	X	ID	2/2	Used
		Description: Code identifying the type of communication number				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. AP: Alternate Phone Number EM: Electronic Mail TE: Telephone Number FX: Fax Number</i>				
		All valid standard codes are used. (Total Codes: 42)				
PER08	364	Communication Number	X	AN	1/256	Used
		Description: Complete communications number including country or area code when applicable				

Syntax Rules:

1. P0304 - If either PER03 or PER04 is present, then the other is required.
2. P0506 - If either PER05 or PER06 is present, then the other is required.
3. P0708 - If either PER07 or PER08 is present, then the other is required.

MAN Marks and Numbers Information

Pos: 3600	Max: >1
Detail - Optional	
Loop: HL	Elements: 6

User Option (Usage): Used

Purpose: To indicate identifying marks and numbers for shipping containers

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
MAN01	88	Marks and Numbers Qualifier Description: Code specifying the application or source of Marks and Numbers (87) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. GM: UCC Serial Shipping Container Code (SSCC) + Application Identifier CP: Carrier-Assigned Package ID Number</i> All valid standard codes are used. (Total Codes: 21)	M	ID	1/2	Must use
MAN02	87	Marks and Numbers Description: Marks and numbers used to identify a shipment or parts of a shipment	M	AN	1/48	Must use
MAN03	87	Marks and Numbers Description: Marks and numbers used to identify a shipment or parts of a shipment	O	AN	1/48	Used
MAN04	88	Marks and Numbers Qualifier Description: Code specifying the application or source of Marks and Numbers (87) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. GM: UCC Serial Shipping Container Code (SSCC) + Application Identifier CP: Carrier-Assigned Package ID Number</i> All valid standard codes are used. (Total Codes: 21)	X	ID	1/2	Used
MAN05	87	Marks and Numbers Description: Marks and numbers used to identify a shipment or parts of a shipment	X	AN	1/48	Used

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
MAN06	87	Marks and Numbers	O	AN	1/48	Used

Description: Marks and numbers used to identify a shipment or parts of a shipment

Syntax Rules:

1. P0405 - If either MAN04 or MAN05 is present, then the other is required.
2. C0605 - If MAN06 is present, then MAN05 is required.

Semantics:

1. MAN01/MAN02 and MAN04/MAN05 may be used to identify two different marks and numbers assigned to the same physical container.
2. When both MAN02 and MAN03 are used, MAN02 is the starting number of a sequential range and MAN03 is the ending number of that range.
3. When both MAN05 and MAN06 are used, MAN05 is the starting number of a sequential range, and MAN06 is the ending number of that range.

Comments:

1. When MAN01 contains code "UC" (U.P.C. Shipping Container Code) and MAN05/MAN06 contain a range of ID numbers, MAN03 is not used. The reason for this is that the U.P.C. Shipping Container code is the same on every carton that is represented in the range in MAN05/MAN06.
2. MAN03 and/or MAN06 are only used when sending a range(s) of ID numbers.
3. When both MAN02/MAN03 and MAN05/MAN06 are used to send ranges of ID numbers, the integrity of the two ID numbers must be maintained.

DTM Date/Time Reference

Pos: 3700	Max: 10
Detail - Optional	
Loop: HL	Elements: 3

User Option (Usage): Used

Purpose: To specify pertinent dates and times

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
DTM01	374	Date/Time Qualifier	M	ID	3/3	Must use
		Description: Code specifying type of date or time, or both date and time SPS Commerce Element Notes: For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. All valid standard codes are used. (Total Codes: 1280)				
DTM02	373	Date	X	DT	8/8	Used
		Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year				
DTM03	337	Time	X	TM	4/8	Used
		Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)				

Syntax Rules:

1. R020305 - At least one of DTM02, DTM03 or DTM05 is required.
2. C0403 - If DTM04 is present, then DTM03 is required.

FOB F.O.B. Related Instructions

Pos: 3800	Max: 1
Detail - Optional	
Loop: HL	Elements: 3

User Option (Usage): Recommended

Purpose: To specify transportation instructions relating to shipment

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
FOB01	146	Shipment Method of Payment	M	ID	2/2	Must use
		Description: Code identifying payment terms for transportation charges SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CC: Collect PP: Prepaid (by seller) DF: Defined by Buyer and Seller All valid standard codes are used. (Total Codes: 31)				
FOB02	309	Location Qualifier	X	ID	1/2	Recommended
		Description: Code identifying type of location SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> OR: Origin DE: Destination All valid standard codes are used. (Total Codes: 184)				
FOB03	352	Description	O	AN	1/80	Recommended
		Description: A free-form description to clarify the related data elements and their content				

Syntax Rules:

1. C0302 - If FOB03 is present, then FOB02 is required.

Semantics:

1. FOB01 indicates which party will pay the carrier.
2. FOB02 is the code specifying transportation responsibility location.
3. FOB06 is the code specifying the title passage location.
4. FOB08 is the code specifying the point at which the risk of loss transfers. This may be different than the location specified in FOB02/FOB03 and FOB06/FOB07.

N1 Party Identification

Pos: 4100	Max: 1
Detail - Optional	
Loop: N1	Elements: 4

User Option (Usage): Recommended

Purpose: To identify a party by type of organization, name, and code

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N101	98	Entity Identifier Code	M	ID	2/3	Must use
Description: Code identifying an organizational entity, a physical location, property or an individual SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below, but a list of qualifiers specific to your trading partners will be provided by your Solution Strategist during the design phase.</i> ST: Ship To SF: Ship From VN: Vendor BT: Bill To All valid standard codes are used. (Total Codes: 1500)						
N102	93	Name	X	AN	1/60	Used
Description: Free-form name						
N103	66	Identification Code Qualifier	X	ID	1/2	Used
Description: Code designating the system/method of code structure used for Identification Code (67) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below, but a list of qualifiers specific to your trading partners will be provided by your Solution Strategist during the design phase.</i> 1: DUNS 9: DUNS + 4 91: Assigned by Seller 92: Assigned by Buyer All valid standard codes are used. (Total Codes: 241)						
N104	67	Identification Code	X	AN	2/80	Used
Description: Code identifying a party or other code						

Syntax Rules:

1. R0203 - At least one of N102 or N103 is required.
2. P0304 - If either N103 or N104 is present, then the other is required.

Comments:

1. This segment, used alone, provides the most efficient method of providing organizational identification. To obtain this efficiency the "ID Code" (N104) must provide a key to the table maintained by the transaction processing party.
2. N105 and N106 further define the type of entity in N101.

SPS Commerce Segment Notes:

For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

N2 Additional Name Information

Pos: 4200	Max: 2
Detail - Optional	
Loop: N1	Elements: 2

User Option (Usage): Used

Purpose: To specify additional names

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N201	93	Name	M	AN	1/60	Must use
		Description: Free-form name				
N202	93	Name	O	AN	1/60	Used
		Description: Free-form name				

N3 Party Location

Pos: 4300	Max: 2
Detail - Optional	
Loop: N1	Elements: 2

User Option (Usage): Recommended

Purpose: To specify the location of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N301	166	Address Information	M	AN	1/55	Must use
		Description: Address information				
N302	166	Address Information	O	AN	1/55	Used
		Description: Address information				

N4 Geographic Location

Pos: 4400	Max: 1
Detail - Optional	
Loop: N1	Elements: 7

User Option (Usage): Recommended

Purpose: To specify the geographic place of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N401	19	City Name	O	AN	2/30	Used
		Description: Free-form text for city name				
N402	156	State or Province Code	X	ID	2/2	Used
		Description: Code (Standard State/Province) as defined by appropriate government agency				
N403	116	Postal Code	O	ID	3/15	Used
		Description: Code defining international postal zone code excluding punctuation and blanks (zip code for United States)				
N404	26	Country Code	X	ID	2/3	Used
		Description: Code identifying the country				
N405	309	Location Qualifier	X	ID	1/2	Used
		Description: Code identifying type of location				
		All valid standard codes are used. (Total Codes: 184)				
N406	310	Location Identifier	O	AN	1/30	Used
		Description: Code which identifies a specific location				
N407	1715	Country Subdivision Code	X	ID	1/3	Used
		Description: Code identifying the country subdivision				

Syntax Rules:

1. E0207 - Only one of N402 or N407 may be present.
2. C0605 - If N406 is present, then N405 is required.
3. C0704 - If N407 is present, then N404 is required.

Comments:

1. A combination of either N401 through N404, or N405 and N406 may be adequate to specify a location.
2. N402 is required only if city name (N401) is in the U.S. or Canada.

HL Hierarchical Level

Pos: 0200	Max: 1
Detail - Mandatory	
Loop: HL	Elements: 3

User Option (Usage): Must use

Purpose: To identify dependencies among and the content of hierarchically related groups of data segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
HL01	628	Hierarchical ID Number	M	AN	1/12	Must use
		Description: A unique number assigned by the sender to identify a particular data segment in a hierarchical structure				
HL02	734	Hierarchical Parent ID Number	O	AN	1/12	Used
		Description: Identification number of the next higher hierarchical data segment that the data segment being described is subordinate to				
HL03	735	Hierarchical Level Code	M	ID	1/2	Must use
		Description: Code defining the characteristic of a level in a hierarchical structure				
		CodeList Summary (Total Codes: 250, Included: 1)				
		<u>Code</u>	<u>Name</u>			
		I	Item			

Comments:

1. The HL segment is used to identify levels of detail information using a hierarchical structure, such as relating line-item data to shipment data, and packaging data to line-item data.
2. The HL segment defines a top-down/left-right ordered structure.
3. HL01 shall contain a unique alphanumeric number for each occurrence of the HL segment in the transaction set. For example, HL01 could be used to indicate the number of occurrences of the HL segment, in which case the value of HL01 would be "1" for the initial HL segment and would be incremented by one in each subsequent HL segment within the transaction.
4. HL02 identifies the hierarchical ID number of the HL segment to which the current HL segment is subordinate.
5. HL03 indicates the context of the series of segments following the current HL segment up to the next occurrence of an HL segment in the transaction. For example, HL03 is used to indicate that subsequent segments in the HL loop form a logical grouping of data referring to shipment, order, or item-level information.
6. HL04 indicates whether or not there are subordinate (or child) HL segments related to the current HL segment.

LIN Item Identification

Pos: 0300	Max: 1
Detail - Mandatory	
Loop: HL	Elements: 20

User Option (Usage): Must use

Purpose: To specify basic item identification data

Element Summary:

Ref	Id	Element Name	Req	Type	Min/Max	Usage
LIN01	350	Assigned Identification	O	AN	1/20	Recommended

Description: Alphanumeric characters assigned for differentiation within a transaction set

LIN02	235	Product/Service ID Qualifier	M	ID	2/2	Must use
-------	-----	------------------------------	---	----	-----	----------

Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234)

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

BP: Buyer Part Number

VN: Vendor Part Number

UP: UPC (12 digit)

UK: GTIN (14 digit)

MG: Manufacturer's Part Number

CB: Buyer's Catalog Number

ZBP: Additional Buyer Part Number

ZVP: Additional Vendor Part Number

**All valid standard codes are used.
(Total Codes: 519)**

LIN03	234	Product/Service ID	M	AN	1/48	Must use
-------	-----	--------------------	---	----	------	----------

Description: Identifying number for a product or service

LIN04	235	Product/Service ID Qualifier	X	ID	2/2	Used
-------	-----	------------------------------	---	----	-----	------

Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234)

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

BP: Buyer Part Number

VN: Vendor Part Number

UP: UPC (12 digit)

UK: GTIN (14 digit)

MG: Manufacturer's Part Number

CB: Buyer's Catalog Number

ZBP: Additional Buyer Part Number

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		<i>ZVP: Additional Vendor Part Number</i>				
		All valid standard codes are used. (Total Codes: 519)				
LIN05	234	Product/Service ID	X	AN	1/48	Used
		Description: Identifying number for a product or service				
LIN06	235	Product/Service ID Qualifier	X	ID	2/2	Used
		Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234)				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. BP: Buyer Part Number VN: Vendor Part Number UP: UPC (12 digit) UK: GTIN (14 digit) MG: Manufacturer's Part Number CB: Buyer's Catalog Number ZBP: Additional Buyer Part Number ZVP: Additional Vendor Part Number</i>				
		All valid standard codes are used. (Total Codes: 519)				
LIN07	234	Product/Service ID	X	AN	1/48	Used
		Description: Identifying number for a product or service				
LIN08	235	Product/Service ID Qualifier	X	ID	2/2	Used
		Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234)				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. BP: Buyer Part Number VN: Vendor Part Number UP: UPC (12 digit) UK: GTIN (14 digit) MG: Manufacturer's Part Number CB: Buyer's Catalog Number ZBP: Additional Buyer Part Number ZVP: Additional Vendor Part Number</i>				
		All valid standard codes are used. (Total Codes: 519)				
LIN09	234	Product/Service ID	X	AN	1/48	Used
		Description: Identifying number for a				

<u>Ref</u>	<u>Id</u>	<u>Element Name</u> product or service	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
LIN10	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>BP: Buyer Part Number</i> <i>VN: Vendor Part Number</i> <i>UP: UPC (12 digit)</i> <i>UK: GTIN (14 digit)</i> <i>MG: Manufacturer's Part Number</i> <i>CB: Buyer's Catalog Number</i> <i>ZBP: Additional Buyer Part Number</i> <i>ZVP: Additional Vendor Part Number</i> All valid standard codes are used. (Total Codes: 519)	X	ID	2/2	Used
LIN11	234	Product/Service ID Description: Identifying number for a product or service	X	AN	1/48	Used
LIN12	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>BP: Buyer Part Number</i> <i>VN: Vendor Part Number</i> <i>UP: UPC (12 digit)</i> <i>UK: GTIN (14 digit)</i> <i>MG: Manufacturer's Part Number</i> <i>CB: Buyer's Catalog Number</i> <i>ZBP: Additional Buyer Part Number</i> <i>ZVP: Additional Vendor Part Number</i> All valid standard codes are used. (Total Codes: 519)	X	ID	2/2	Used
LIN13	234	Product/Service ID Description: Identifying number for a product or service	X	AN	1/48	Used
LIN14	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234)	X	ID	2/2	Used

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> BP: Buyer Part Number VN: Vendor Part Number UP: UPC (12 digit) UK: GTIN (14 digit) MG: Manufacturer's Part Number CB: Buyer's Catalog Number ZBP: Additional Buyer Part Number ZVP: Additional Vendor Part Number All valid standard codes are used. (Total Codes: 519)				
LIN15	234	Product/Service ID Description: Identifying number for a product or service	X	AN	1/48	Used
LIN16	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> BP: Buyer Part Number VN: Vendor Part Number UP: UPC (12 digit) UK: GTIN (14 digit) MG: Manufacturer's Part Number CB: Buyer's Catalog Number ZBP: Additional Buyer Part Number ZVP: Additional Vendor Part Number All valid standard codes are used. (Total Codes: 519)	X	ID	2/2	Used
LIN17	234	Product/Service ID Description: Identifying number for a product or service	X	AN	1/48	Used
LIN18	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234) SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> BP: Buyer Part Number VN: Vendor Part Number	X	ID	2/2	Used

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		UP: UPC (12 digit) UK: GTIN (14 digit) MG: Manufacturer's Part Number CB: Buyer's Catalog Number ZBP: Additional Buyer Part Number ZVP: Additional Vendor Part Number All valid standard codes are used. (Total Codes: 519)				
LIN19	234	Product/Service ID	X	AN	1/48	Used
		Description: Identifying number for a product or service				
LIN27	234	Product/Service ID	X	AN	1/48	Used
		Description: Identifying number for a product or service				

Semantics:

1. LIN01 is the line item identification

Comments:

1. See the Data Dictionary for a complete list of IDs.
2. LIN02 through LIN31 provide for fifteen different product/service IDs for each item. For example: Case, Color, Drawing No., U.P.C. No., ISBN No., Model No., or SKU.

SN1 Item Detail (Shipment)

Pos: 0400	Max: 1
Detail - Mandatory	
Loop: HL	Elements: 7

User Option (Usage): Must use

Purpose: To specify line-item detail relative to shipment

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
SN101	350	Assigned Identification Description: Alphanumeric characters assigned for differentiation within a transaction set	O	AN	1/20	Recommended
SN102	382	Number of Units Shipped Description: Numeric value of units shipped in manufacturer's shipping units for a line item or transaction set	M	R	1/10	Must use
SN103	355	Unit or Basis for Measurement Code Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CA: Case DZ: Dozen EA: Each UN: Unit All valid standard codes are used. (Total Codes: 844)	M	ID	2/2	Must use
SN104	646	Quantity Shipped to Date Description: Number of units shipped to date	O	R	1/15	Used
SN105	380	Quantity Description: Numeric value of quantity	O	R	1/15	Used
SN106	355	Unit or Basis for Measurement Code Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CA: Case	X	ID	2/2	Used

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		<i>DZ: Dozen</i> <i>EA: Each</i> <i>UN: Unit</i>				
		All valid standard codes are used. (Total Codes: 844)				
SN108	668	Line Item Status Code	O	ID	2/2	Used
		Description: Code specifying the action taken by the seller on a line item requested by the buyer				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>IA: Item Accepted</i> <i>BP: Item Accepted - Partial Shipment, Balance Backordered</i>				
		All valid standard codes are used. (Total Codes: 26)				

Syntax Rules:

1. P0506 - If either SN105 or SN106 is present, then the other is required.

Semantics:

1. SN101 is the ship notice line-item identification.
2. SN105 is quantity ordered.

Comments:

1. SN103 defines the unit of measurement for both SN102 and SN104.

PO4 Item Physical Details

Pos: 0700	Max: 1
Detail - Optional	
Loop: HL	Elements: 14

User Option (Usage): Recommended

Purpose: To specify the physical qualities, packaging, weights, and dimensions relating to the item

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
PO401	356	Pack	O	N0	1/6	Recommended
		Description: The number of inner containers, or number of eaches if there are no inner containers, per outer container				
PO402	357	Size	X	R	1/8	Used
		Description: Size of supplier units in pack				
PO403	355	Unit or Basis for Measurement Code	X	ID	2/2	Used
		Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CA: Case EA: Each				
		All valid standard codes are used. (Total Codes: 844)				
PO404	103	Packaging Code	X	AN	3/5	Used
		Description: Code identifying the type of packaging; Part 1: Packaging Form, Part 2: Packaging Material; if the Data Element is used, then Part 1 is always required				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> CTN: Carton PLT: Pallet 25: Corrugated 94: Wood				
		All valid standard codes are used. (Total Codes: 212)				
PO405	187	Weight Qualifier	O	ID	1/2	Used
		Description: Code defining the type of weight				

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>G: Gross Weight</i> <i>N: Actual Net Weight</i>				
		All valid standard codes are used. (Total Codes: 52)				
PO406	384	Gross Weight per Pack	X	R	1/9	Used
		Description: Numeric value of gross weight per pack				
PO407	355	Unit or Basis for Measurement Code	X	ID	2/2	Used
		Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>LB: Pound</i> <i>KG: Kilogram</i>				
		All valid standard codes are used. (Total Codes: 844)				
PO408	385	Gross Volume per Pack	X	R	1/9	Used
		Description: Numeric value of gross volume per pack				
PO409	355	Unit or Basis for Measurement Code	X	ID	2/2	Used
		Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>CF: Cubic Feet</i>				
		All valid standard codes are used. (Total Codes: 844)				
PO410	82	Length	X	R	1/8	Used
		Description: Largest horizontal dimension of an object measured when the object is in the upright position				
PO411	189	Width	X	R	1/8	Used
		Description: Shorter measurement of the				

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		two horizontal dimensions measured with the object in the upright position				
PO412	65	Height	X	R	1/8	Used
		Description: Vertical dimension of an object measured when the object is in the upright position				
PO413	355	Unit or Basis for Measurement Code	X	ID	2/2	Used
		Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken				
		SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst. CF: Cubic Feet</i>				
		All valid standard codes are used. (Total Codes: 844)				
PO414	810	Inner Pack	O	N0	1/6	Recommended
		Description: The number of eaches per inner container				

Syntax Rules:

1. PO203 - If either PO402 or PO403 is present, then the other is required.
2. C0506 - If PO405 is present, then PO406 is required.
3. P0607 - If either PO406 or PO407 is present, then the other is required.
4. P0809 - If either PO408 or PO409 is present, then the other is required.
5. C1013 - If PO410 is present, then PO413 is required.
6. C1113 - If PO411 is present, then PO413 is required.
7. C1213 - If PO412 is present, then PO413 is required.
8. L13101112 - If PO413 is present, then at least one of PO410, PO411 or PO412 is required.

Semantics:

1. PO415 is used to indicate the relative layer of this package or range of packages within the layers of packaging. Relative Position 1 (value R1) is the innermost package.
2. PO416 is the package identifier or the beginning package identifier in a range of identifiers.
3. PO417 is the ending package identifier in a range of identifiers.
4. PO418 is the number of packages in this layer.

Comments:

1. PO403 - The "Unit or Basis for Measure Code" in this segment position is for purposes of defining the unit of measure of the "Size" identified in the PO402. For example: If the carton contains 24 12-Ounce packages, it would be described as follows: Data element 356 = "24"; Data element 357 = "12"; Data element 355 = "OZ".
2. PO413 defines the unit of measure for PO410, PO411, and PO412.

PID Product/Item Description

Pos: 0800	Max: 200
Detail - Optional	
Loop: HL	Elements: 5

User Option (Usage): Recommended

Purpose: To describe a product or process in coded or free-form format

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
PID01	349	Item Description Type	M	ID	1/1	Must use
		Description: Code indicating the format of a description SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>F: Free Form</i> All valid standard codes are used. (Total Codes: 3)				
PID02	750	Product/Process Characteristic Code	O	ID	2/3	Recommended
		Description: Code identifying the general class of a product or process characteristic SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>08: Product Description</i> All valid standard codes are used. (Total Codes: 238)				
PID03	559	Agency Qualifier Code	X	ID	2/2	Used
		Description: Code identifying the agency assigning the code values SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>VI: Voluntary Inter-Industry Commerce Standard (VICS) EDI</i> All valid standard codes are used. (Total Codes: 194)				
PID04	751	Product Description Code	X	AN	1/12	Used
		Description: A code from an industry code list which provides specific data about a product characteristic SPS Commerce Element Notes: <i>This element typically used for noting</i>				

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		<i>Compliance with Fair Labor Standards Act (DOLFL)</i>				
PID05	352	Description	X	AN	1/80	Used
		Description: A free-form description to clarify the related data elements and their content				

Syntax Rules:

1. C0403 - If PID04 is present, then PID03 is required.
2. R0405 - At least one of PID04 or PID05 is required.

Semantics:

1. Use PID03 to indicate the organization that publishes the code list being referred to.
2. PID04 should be used for industry-specific product description codes.
3. PID08 describes the physical characteristics of the product identified in PID04. A "Y" indicates that the specified attribute applies to this item; an "N" indicates it does not apply. Any other value is indeterminate.
4. PID09 is used to identify the language being used in PID05.

Comments:

1. If PID01 equals "F", then PID05 is used. If PID01 equals "S", then PID04 is used. If PID01 equals "X", then both PID04 and PID05 are used.
2. Use PID06 when necessary to refer to the product surface or layer being described in the segment.
3. PID07 specifies the individual code list of the agency specified in PID03.

REF Reference Information

Pos: 1900	Max: >1
Detail - Optional	
Loop: HL	Elements: 2

User Option (Usage): Recommended

Purpose: To specify identifying information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
REF01	128	Reference Identification Qualifier	M	ID	2/3	Must use
		Description: Code qualifying the Reference Identification SPS Commerce Element Notes: <i>Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.</i> <i>SE: Serial Number</i> <i>LT: Lot Number</i> All valid standard codes are used. (Total Codes: 1731)				
REF02	127	Reference Identification	X	AN	1/50	Used
		Description: Reference information as defined for a particular Transaction Set or as specified by the Reference Identification Qualifier				

Syntax Rules:

1. R0203 - At least one of REF02 or REF03 is required.

Semantics:

1. REF04 contains data relating to the value cited in REF02.

DTM Date/Time Reference

Pos: 3700	Max: 10
Detail - Optional	
Loop: HL	Elements: 3

User Option (Usage): Used

Purpose: To specify pertinent dates and times

Element Summary:

Ref	Id	Element Name	Req	Type	Min/Max	Usage
DTM01	374	Date/Time Qualifier	M	ID	3/3	Must use

Description: Code specifying type of date or time, or both date and time

SPS Commerce Element Notes:

Commonly used qualifiers are listed below. For a definitive list of qualifiers specific to your trading partners, please reach out to your Business Analyst.

001: Cancel Date

002: Requested Delivery Date

010: Requested Ship Date

011: Actual Ship Date

067: Current Schedule Delivery Date

405: Production

511: Shelf Life Expiration

**All valid standard codes are used.
(Total Codes: 1280)**

DTM02	373	Date	X	DT	8/8	Used
-------	-----	------	---	----	-----	------

Description: Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year

DTM03	337	Time	X	TM	4/8	Used
-------	-----	------	---	----	-----	------

Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)

Syntax Rules:

1. R020305 - At least one of DTM02, DTM03 or DTM05 is required.

CTT Transaction Totals

Pos: 0100	Max: 1
Summary - Optional	
Loop: N/A	Elements: 2

User Option (Usage): Used

Purpose: To transmit a hash total for a specific element in the transaction set

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
CTT01	354	Number of Line Items	M	N0	1/6	Must use
		Description: Total number of line items in the transaction set				
CTT02	347	Hash Total	O	R	1/10	Used
		Description: Sum of values of the specified data element. All values in the data element will be summed without regard to decimal points (explicit or implicit) or signs. Truncation will occur on the left most digits if the sum is greater than the maximum size of the hash total of the data element. Example: -.0018 First occurrence of value being hashed. .18 Second occurrence of value being hashed. 1.8 Third occurrence of value being hashed. 18.01 Fourth occurrence of value being hashed. 18E2 Fifth occurrence of value being hashed. ----- 1873 Hash Total				

Syntax Rules:

1. P0304 - If either CTT03 or CTT04 is present, then the other is required.
2. P0506 - If either CTT05 or CTT06 is present, then the other is required.

Comments:

1. This segment is intended to provide hash totals to validate transaction completeness and correctness.

SE Transaction Set Trailer

Pos: 0200	Max: 1
Summary - Mandatory	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To indicate the end of the transaction set and provide the count of the transmitted segments (including the beginning (ST) and ending (SE) segments)

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
SE01	96	Number of Included Segments	M	N0	1/10	Must use
		Description: Total number of segments included in a transaction set including ST and SE segments				
SE02	329	Transaction Set Control Number	M	AN	4/9	Must use
		Description: Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set				

Comments:

1. SE is the last segment of each transaction set.

GE Functional Group Trailer

Pos: 0100	Max: 1
Not Defined - Mandatory	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To indicate the end of a functional group and to provide control information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
GE01	97	Number of Transaction Sets Included	M	N0	1/6	Must use
		Description: Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element				
GE02	28	Group Control Number	M	N0	1/9	Must use
		Description: Assigned number originated and maintained by the sender				

Semantics:

1. The data interchange control number GE02 in this trailer must be identical to the same data element in the associated functional group header, GS06.

Comments:

1. The use of identical data interchange control numbers in the associated functional group header and trailer is designed to maximize functional group integrity. The control number is the same as that used in the corresponding header.

IEA Interchange Control Trailer

Pos: 0200	Max: 1
Not Defined - Mandatory	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To define the end of an interchange of zero or more functional groups and interchange-related control segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
IEA01	I16	Number of Included Functional Groups	M	N0	1/5	Must use
		Description: A count of the number of functional groups included in an interchange				
IEA02	I12	Interchange Control Number	M	N0	9/9	Must use
		Description: A control number assigned by the interchange sender				